AI and Digital Transformation: Impact on Business Practices, Firm Performance, and Corporate Reporting

A Structured Research Brief, 2015-2025

Abstract

This research brief synthesises evidence on the impact of artificial intelligence (AI) and digital transformation on business practices, firm performance, and corporate reporting over the past decade. Drawing on OECD statistics, industry surveys from McKinsey, Deloitte, PwC, and EY, peer-reviewed academic research, and regulatory developments from the EU, SEC, and standard-setting bodies, the brief identifies adoption trends, productivity effects, audit transformation, disclosure implications, and research opportunities for accounting scholars.

1. AI and Digital Technology Adoption Trends

AI adoption is highly concentrated in knowledge-intensive services. The ICT sector leads with 57.3% of firms using AI in 2025, followed by professional, scientific, and technical services at 36.8%. In 2024, the most recent year with full sectoral coverage, ICT adoption stood at 44.6%, compared with 7.2% in construction and 7.8% in accommodation and food services (OECD, 2026). The pattern reflects both the origin of AI technologies in the ICT sector and the relative ease of integrating AI into digital-first workflows. Among the lagging sectors, 2025 growth was strongest in accommodation and food services (62.5% year-on-year) and construction (59.1%), suggesting catch-up is underway (OECD, 2026).

Firm size remains the strongest predictor of AI adoption. In 2024, 40% of firms with 250 or more employees actively used AI, compared with 11.5% of SMEs. The adoption gap has widened: in 2020, large firms were 4.3 times more likely to use AI than SMEs; by 2024, this ratio stood at 3.5 times, but the absolute gap grew from 13.8 to 28.5 percentage points. Among SMEs that use generative AI, only 29% deploy it in core business activities; the majority confine AI to peripheral tasks. Younger firms, including startups, show notably higher AI adoption regardless of scale, suggesting that organisational inertia, rather than resource constraints alone, drives the gap (OECD, 2025). Preliminary 2025 data from the OECD show large-firm adoption reaching 52.0% and small-firm adoption at 17.4%, indicating the gap continues to expand (OECD, 2026).

OECD data reveal a structural break in 2024 coinciding with the widespread availability of general-purpose generative AI tools such as ChatGPT and Copilot. Between 2023 and 2025, aggregate AI adoption more than doubled from 8.7% to 20.2%. McKinsey's broader global survey (which counts any use in at least one business function) places the 2025 figure at 88%. The divergence between OECD official statistics (firm-level, 10+ employees, strict definition) and McKinsey survey data (any functional use) highlights definitional challenges in measuring AI adoption (OECD, 2026; McKinsey, 2025).

1.1 Adoption by Sector

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1.2 Adoption by Firm Size

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1.3 Adoption Over Time and the GenAI Acceleration

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2. Effects on Firm Productivity, Cost Structures, and Financial Performance

The academic and industry evidence on AI's productivity effects is broadly positive but mixed. OECD estimates suggest AI could add 0.2 to 1.3 percentage points in annual labour productivity growth across G7 economies over the next decade. At the firm level, AI users show productivity premiums of 4% to 15%, depending on the study and methodology (OECD, 2025). Productivity gains may follow a J-shaped curve: declining temporarily before improving, as firms undergo costly reorganisation to integrate AI into their workflows.

McKinsey identified 46 "GenAI high performers" among 876 surveyed firms (5.3%). These leaders attribute over 10% of EBIT to AI deployment and achieve returns exceeding $10.30 per dollar invested, nearly three times the average. Yet the contrast with PwC's finding that 56% of CEOs report zero measurable AI ROI in the past 12 months underscores the implementation gap. Only 6% of organisations have achieved significant enterprise-wide AI impact (McKinsey, 2025; PwC, 2026).

3. Digital Transformation in Auditing, Financial Reporting, and Management Control

The Big Four accounting firms have invested heavily in AI platforms that now analyse entire populations of journal entries rather than traditional samples. EY's Helix platform analyses 100% of client journal entries. PwC developed GL.ai with H2O.ai to detect irregularities invisible to traditional sampling. KPMG's Ignite platform scans millions of accounting entries using machine learning. Deloitte's Zora AI, built with Nvidia, automates finance and procurement workflows, with projected cost reductions of up to 25% in these functions (PwC, 2023; KPMG, 2024; Deloitte, 2025).

KPMG committed USD 2 billion over five years (2020-2025) targeting USD 12 billion in added revenue. PwC invested USD 1 billion in generative AI through its partnership with Microsoft and OpenAI. Adoption of AI-assisted tax preparation surged, with some firms reporting over 80% of individual return preparation handled through automated workflows in 2025 (KPMG, 2024; PwC, 2023).

Figure 5: Big Four AI Audit Platforms, Capabilities, and Investments

Firm	AI Platform	Key Capability	Disclosed Investment
Deloitte	Zora AI (with Nvidia)	Finance and procurement automation; projected 25% cost reduction	Not separately disclosed
PwC	GL.ai (with H2O.ai)	General ledger anomaly detection; partnership with Microsoft/OpenAI	USD 1 billion (2023)
EY	EY Helix; EY Atlas	100% journal entry population analysis; global knowledge platform	Not separately disclosed
KPMG	KPMG Ignite; Clara; Workbench	ML-driven anomaly scanning; real-time audit insights; multi-agent approach	USD 2 billion (2020-2025)

Note: Source: PwC (2023); KPMG (2024); Deloitte (2025); EY (2024).

AI is reshaping management accounting from a reactive, backward-looking function toward proactive, strategic partnership. Machine learning applications in cost control, budgeting, and performance measurement enable predictive and prescriptive analytics that simulate complex business scenarios (Ranta, Ylinen, and Jarvenpaa, 2023). However, Wassie and Lakatos (2024), reviewing 62 articles published between 2019 and 2023, found that Asia and Europe dominate AI accounting research, with the Middle East and Africa showing minimal engagement. The literature reveals critical gaps in governance frameworks, empirical validation, and skill development pathways.

> *Key development: The FRC (UK) raised concerns in June 2025 about the rising use of AI by auditors, questioning whether audit quality controls keep pace with AI deployment. The*

> *PCAOB similarly highlighted AI in its 2024 inspection reports, noting persistent audit deficiencies at firms despite AI adoption.*

3.1 The Big Four and AI-Driven Audit Transformation

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3.2 Management Control Systems and Digital Transformation

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4. Country-Level Digital Readiness and Regulatory Developments

Singapore, Switzerland, and Denmark lead the IMD World Digital Competitiveness Ranking (2024), measured across knowledge, technology, and future readiness pillars (IMD, 2024). Within the EU, Finland (76.0), the Netherlands (72.5), and Denmark (71.0) score highest on the DESI 2024 composite index (European Commission, 2024). The World Bank's Digital Adoption Index, though now dated (2016 for most countries), provides a useful baseline, with a 0-1 scale showing advanced economies clustered above 0.75 and emerging economies substantially lower, with India at 0.42 (World Bank, 2016).

The EU AI Act, adopted in March 2024 and entering into force on 1 August 2024, is the world's first comprehensive AI regulation. It uses a risk-based framework with four categories (unacceptable, high, limited, minimal), phased compliance deadlines through 2030, and penalties up to 7% of global annual turnover for non-compliance. Prohibited practices provisions applied from February 2025; high-risk system obligations begin phasing in from August 2026.

In the United States, the SEC's Investor Advisory Committee recommended AI-specific disclosure guidelines in December 2025, and S&P 500 companies have significantly expanded AI risk factor disclosures. Reputational risk from AI is the most frequently cited concern (38% of S&P 500 firms in 2025), followed by cybersecurity risk (20%) and regulatory uncertainty (41 firms explicitly flag the EU AI Act). AI-related securities class actions rose from 7 cases in 2023 to 14 in 2024 (Conference Board/ESGAUGE, 2025; Fisher & Phillips, 2025).

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4.2 Regulatory Landscape

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5. Corporate AI/Digital Investment Disclosures and Capital Market Implications

AI-related disclosures in corporate filings have grown rapidly. The Conference Board and ESGAUGE report that 72% of S&P 500 companies disclosed at least one material AI risk in their 2025 10-K filings, up from just 12% in 2023 (Conference Board/ESGAUGE, 2025). Eisfeldt et al. (2023) find that firms with higher exposure to generative AI experienced larger increases in market value following the release of ChatGPT. Earnings call analysis by the European Central Bank confirms that early AI engagement boosted stock market performance beyond the immediate impact on expected earnings (Lopardo et al., 2024).

Babina et al. (2024) show that AI-investing firms experience higher growth in sales, employment, and market valuations, driven primarily by product innovation. These findings underscore that AI disclosures convey economically meaningful information to capital markets, but also raise questions about whether firms with limited AI substance are using disclosure language to ride the AI narrative without commensurate investment.

6. Research Gaps and Opportunities for Accounting Scholars

The intersection of AI, digital transformation, and accounting presents substantial opportunities for high-impact research. The following table identifies specific gaps and relevant journal outlets for accounting scholars.

Research Gap	Potential Contribution	Relevant Journals
AI disclosure quality and market pricing	Develop and validate a measure of substantive vs. opportunistic AI disclosure; test whether auditors or regulators can improve disclosure credibility.	The Accounting Review, Journal of Accounting Research, Journal of Accounting and Economics

Research Gap	Potential Contribution	Relevant Journals
AI in audit methodology and audit quality	Examine whether AI-assisted audits reduce restatements, improve fraud detection, or alter auditor judgment. Natural experiments from Big Four AI rollouts.	Auditing: A Journal of Practice and Theory, Contemporary Accounting Research
Management control in AI-intensive firms	How do AI predictions alter budgeting, performance evaluation, and incentive design? Field studies in firms transitioning to AI-driven decision-making.	Management Science, Accounting, Organizations and Society
AI adoption and cost stickiness	Does AI investment change firms' cost behaviour, such as asymmetric cost responses to revenue changes? Links to the cost stickiness literature.	The Accounting Review, Review of Accounting Studies
Digital reporting standards and comparability	Assess the impact of the IASB's digital financial reporting project on cross-firm and cross-country comparability. Inline XBRL adoption effects.	European Accounting Review, Accounting and Business Research
AI governance and internal control	How do firms design internal controls over AI systems? Implications for SOX 404 and internal control frameworks in the AI era.	Journal of Accounting Research, Contemporary Accounting Research
Cross-country digital readiness and firm outcomes	How does country-level digital infrastructure moderate the relationship between firm-level AI investment and performance?	Journal of International Business Studies, Journal of Accounting and Economics
AI, labour, and human capital disclosures	Extend Eisfeldt et al. (2023): how do AI workforce investments interact with financial reporting and voluntary disclosure choices?	Journal of Financial Economics, The Accounting Review

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